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Trimming Tools in Exploratory Data Analysis

Luis Angel García-Escudero, Alfonso Gordaliza, and Carlos Matrán

Exploratory graphical tools based on trimming are proposed for detecting main clusters in a given dataset. The trimming is obtained by resorting to trimmed k-means methodology. The analysis always reduces to the examination of real valued curves, even in the multivariate case. As the technique is based on a robust clustering criterium, it is able to handle the presence of different kinds of outliers. An algorithm is proposed to carry out this (computer intensive) method. As with classical k-means, the method is specially oriented to mixtures of spherical distributions. A possible generalization is outlined to overcome this drawback.

Key Words: Cluster analysis; k-means; Outlier; Robustness; Trimmed k-means.

1. INTRODUCTION

Cluster analysis and outlier detection are closely related problems. This claim has a wider justification in Rocke and Woodruff (1996, 1999). Keeping in mind this idea, the main interest of this article is to propose graphical techniques that help us to distinguish between the *bulk of the data* (those observations following the pattern of the majority) and the *outlying observations.* Additionally, we will be interested in discovering the number of groups or clusters constituting that bulk of the data.

Often, the proposed method also allows to differentiate between *outlying clusters* and *radial outliers.* The first kind of outliers is made up of groups of outliers that differ from the "proper" clusters, considered in the bulk of the data, in that their size is considerably smaller (they do not have enough "strength" to be considered as main clusters). The radial outliers are isolated outliers, each forming its own group.

The determination of the number of groups in the data (say k) is a problem widely treated in the literature (see, e.g., Milligan and Cooper 1985 for a comparative review). The problem is also related to mode assessment or modality problems (Good and Gaskins 1980; Müller and Sawitzki 1987; and Izenman and Sommer 1989 are some references on this topic), but our approach will undoubtedly have more clustering flavor (specially with

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k-means clustering) than with the classical definition of mode as a local maximum of the density.

To introduce the method let us start by recalling the well-known k-means problem: *Given observations $X_1, X_2, \ldots, X_n$, we search for k points (centers), $m_1, \ldots, m_k$, where the minimum in the following expression is attained*

$$W_k := \min_{m_1,\ldots,m_k} \frac{1}{n} \sum_{i=1}^{n} \inf_{1\le j\le k} \|X_i - m_j\|^2 . \tag{1.1}$$

These k centers induce a partition of the given dataset into k groups in an obvious way. W_k is a measure of the variance "within" groups and we will refer to it as the k-variance.

One of the main drawbacks of the k-means methodology is that the number k must be fixed in advance. Many attempts to (self-) determine the optimal k are based on the study of the size of W_k. However, these approaches are easily affected by the lack of robustness of W_k. As an example, Hartigan (1978) and Engelman and Hartigan (1969) provided a multimodality test based on the maximum F-ratio for differences between groups or, equivalently, on the magnitude of W_k. That test unfortunately wrongly decides with high probability that a long-tailed unimodal distribution has more than one mode.

Throughout this article we will resort to "trimmed versions" of W_k trying to avoid the lack of robustness and providing a dynamic way of analyzing data through the consideration of different trimming levels. The question of interest is how to properly trim the "within groups" variance W_k. The problem is particularly hard in the multivariate case, where no privileged directions for removing data can be chosen. In order to solve this difficulty we will follow the so-called, "impartial trimming" procedure (Gordaliza 1991; Cuesta-Albertos, Gordaliza, and Matrán 1997). The key idea is that the data should tell us which are the regions of the sample space to be trimmed.

Definition 1. *(Trimmed k-Variance): For a trimming size α and a distribution F on $\mathbb{R}^p$, the population trimmed k-variance, $W_k(\alpha)$, is defined through the constrained double minimization procedure*

$$W_k(\alpha) := \min_{B:F(B)\ge 1-\alpha} \quad \min_{\{m_1,\ldots,m_k\}\subset\mathbb{R}^p} \frac{1}{F(B)} \int_B \inf_{i=1,\ldots,k} \|x - m_i\|^2 \, dF(x), \tag{1.2}$$

where B could be any Borel set in $\mathbb{R}^p$.

The set with k points and the set B where the minimum of the previous expression is attained will be a trimmed k-mean and an optimal set, respectively. Cuesta-Albertos et al. (1997) showed the important fact that the optimal set B can be always taken (essentially) as the union of k balls with equal radius and centred on the trimmed k-means.

The sample analogue to the problem stated in (1.2) is obtained as follows:

Definition 2. *(Empirical Trimmed k-Variance): Given observations $X_1, \ldots, X_n$ sampled from a distribution F, the empirical trimmed k-variance is defined as*

$$\widehat{W_k}(\alpha) := \min_{\mathbf{Y}} \quad \min_{\{m_1,\ldots,m_k\}\subset\mathbb{R}^p} \frac{1}{\lceil n(1-\alpha)\rceil} \sum_{X_j\in\mathbf{Y}} \inf_{i=1,\ldots,k} \|X_j - m_i\|^2 , \tag{1.3}$$

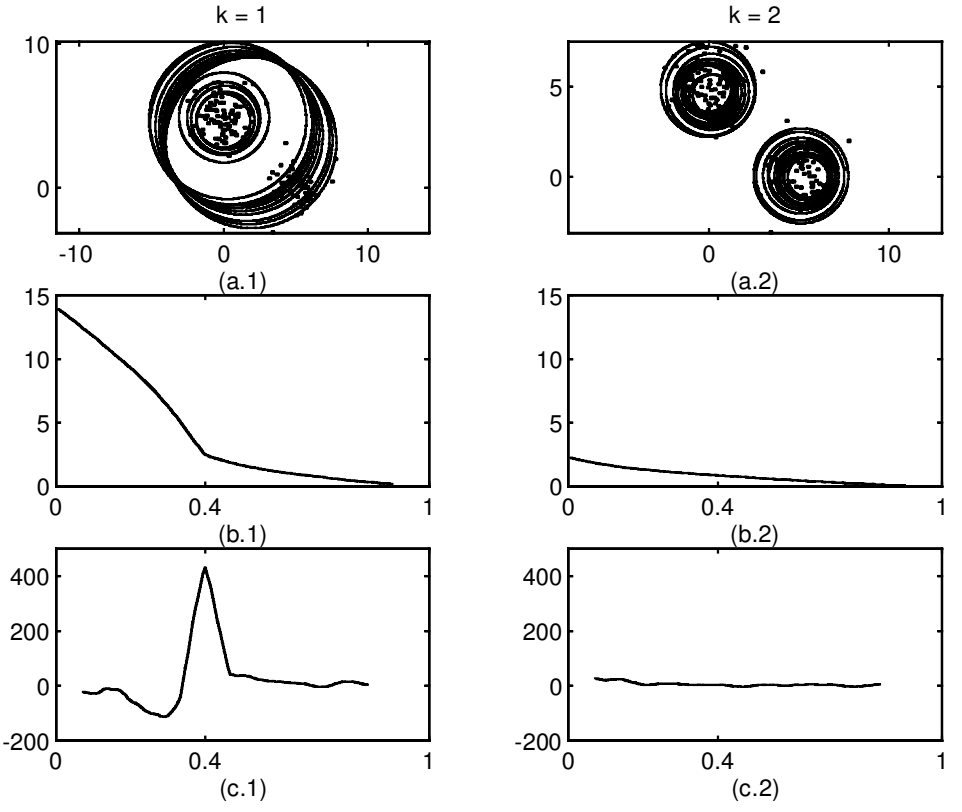


Figure 1. A simulated dataset and the sequences of empirical optimal sets when $k = 1$ (a.1) and $k = 2$ (a.2). Curves $\widehat{W_1}$ (b.1) and $\widehat{W_2}$ (b.2). (c.1) and (c.2) are the numerical second derivatives.

where $\mathbf{Y}$ *ranges on the set of subsets of* $\{X_1, \ldots, X_n\}$ *containing* $\lceil n(1-\alpha) \rceil$ *data points* ($\lceil z \rceil$ *denotes the smallest integer rounding* z *by excess).*

Now, this problem leads to the empirical trimmed k-means and the empirical optimal sets (i.e., sets containing the nontrimmed observations). In the univariate case, the solution for (1.3) when $k = 1$ coincides with Rousseeuw's (1985) least trimmed squares (LTS) location estimator. More details covering different aspects of trimmed k-means can be found in Cuesta-Albertos et al. (1997), García-Escudero, Gordaliza, and Matrán (1999a,b) and García-Escudero and Gordaliza (1999).

Let us begin considering a simple example to illustrate the main idea underlying the proposed method.

Example 1. We have generated a bivariate dataset containing 150 observations from which 90 observations (60%) came from a $N(\mu_1, I)$ and 60 (40%) from a $N(\mu_2, I)$, where $\mu_1 = (0, 5)'$, $\mu_2 = (5, 0)'$ and I denotes the identity matrix in $\mathbb{R}^2$. Figure 1 shows a sequence of optimal sets associated with the empirical trimmed 1- and 2-mean for different trimming sizes. Optimal sets contain the nontrimmed observations (so, the greater the trimming size the less the number of data points within the optimal set).

For a trimming size α, let us measure the "within groups" dispersion *restricted to the corresponding optimal sets,* when $k = 1$ and 2, through $\widehat{W}_1(\alpha)$ and $\widehat{W}_2(\alpha)$. If we plot them against α, we obtain the curves in Figure 1.b's, that will be called the (empirical) trimmed 1- and 2-variance functionals. When an improper choice of k has been made ($k = 1$ in this case), we will have an "overdispersion" phenomenon inside the optimal sets for some trimming sizes. So, as we can see in Figure 1.b.1, the curve decreases very fast from $\alpha = 0$ to 0.4 while points belonging to the small cluster are being trimmed off. These points are placed far from the center of the main cluster and have a high contribution to the trimmed "within groups" variance. After $\alpha = 0.4$, a softer decrease begins because the trimming size allows us to delete completely the small cluster and the optimal set has now a one cluster structure. To emphasize these changes in the rates of decrease, we have plotted a numerical second derivative of those curves in Figure 1.c's. Note that a clear peak appears at $\alpha = 0.4$ in Figure 1.c.1.

From this example, we can guess some applications that the careful analysis of the trimmed k-variance functionals could provide:

- *Guidance for choosing the right k:* The smallest k such that the rate of decrease of the trimmed k-variance is smooth over all the range of trimming sizes could be an appropriate choice for k.
- *Information about the size of the clusters:* Abrupt changes in the pattern of decrease are associated with trimming sizes which exclude complete clusters.
- *Protection against different kinds of contamination:* Since trimmed k-means were initially intended for doing robust clustering, the procedure will be able to handle different kinds of contamination (García-Escudero and Gordaliza 1999).

It is important to realize that our approach reduces to the examination of real-valued curves whatever the dimension of the sample space. Notice also that, as this approach does not have any "local" character, it will be less affected by "curse of dimensionality" troubles. Other advantages of this approach arise from the fact that we can also retain the location of the centers and optimal sets to be used for other statistical purposes. Additionally, these functionals may be seen as "concentration" measures admitting a similar treatment as other ones based on different "depth" concepts (e.g., Liu, Parelius, and Singh 1999). Let us comment, finally, that another interesting approach that allows us to explore the clustering and outlier structure from a "reachability" viewpoint appears in Ankerst, Breunig, Kriegel, and Sander (1999). The procedure is based on a more local definition of outlier (more references on this approach can be found at http://www.dbs.informatik.uni-muenchen.de).

Our procedure inherits the computational complexity of trimmed k-means. So, as part of the core of this article, we provide in Section 3 feasible algorithms to carry out this approach.

Provided that the optimal zones are always a union of balls, the procedure is particularly well suited to mixtures of spherical distributions. This could be sometimes considered as a serious drawback, because other more general mixtures and ways of measuring distances for elements in a given group appear in practice. However, a generalization to cover these possibilities will be given in Section 4.

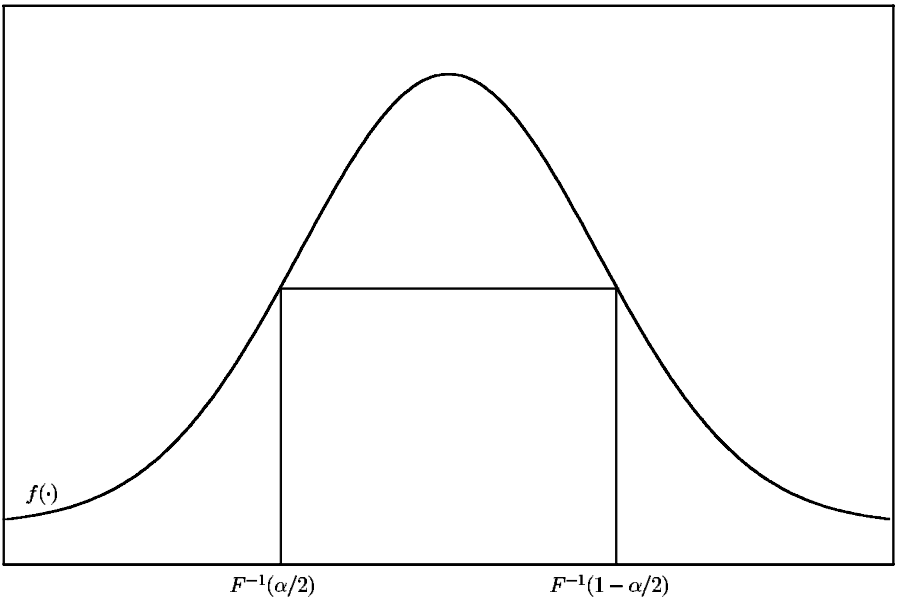


Figure 2. Inequality (2.1) holds for unimodal symmetric distributions.

2. TRIMMED k-VARIANCE FUNCTIONALS

The basic tool in this approach is the study of *trimmed k-variance functionals* defined as

$$W_k : \alpha \mapsto W_k(\alpha), \quad k = 1, 2, \ldots,$$

where $W_k(\alpha)$ was defined in (1.2). Some properties of these functionals are:

(i) Values of curves at $\alpha = 0$ coincide with the k-variances, W_k's, in (1.1).
(ii) The functionals are continuous for absolutely continuous distributions.
(iii) They are monotonically decreasing in α (Lemma 2.2 in Cuesta-Albertos et al. 1997).
(iv) For fixed α, $W_k(\alpha)$ decreases when k increases. Moreover, $W_i(\alpha)$ is strictly greater than $W_j(\alpha)$ for $j > i$ whenever $W_i(\alpha) > 0$ (Prop. A.2 in Cuesta-Albertos et al. 1997).

Property (iii) entails the decreasing character of W_k, but the rate of decrease can be very different depending on k. For instance, when a "proper" k (for the number of main clusters) has been chosen, the rate of decrease of W_k is similar to that obtained by working individually in each population cluster with W_1. In this case, the decrease is smooth and no sign changes in the second derivative should be expected.

Examples given later in this section will serve to clarify these key assertions from the sample viewpoint; however, let us begin by analyzing how the decrease of $W_1(\cdot)$ should be in a very simple situation. Assume that F is the distribution function of a continuous real-valued random variable with symmetric unimodal density f. Under these conditions, the trimmed 1-mean coincides with the symmetry center of the distribution and the optimal set is the interval $[F^{-1}(\alpha/2), F^{-1}(1-\alpha/2)]$ (the midpoint of this interval is also the symmetry

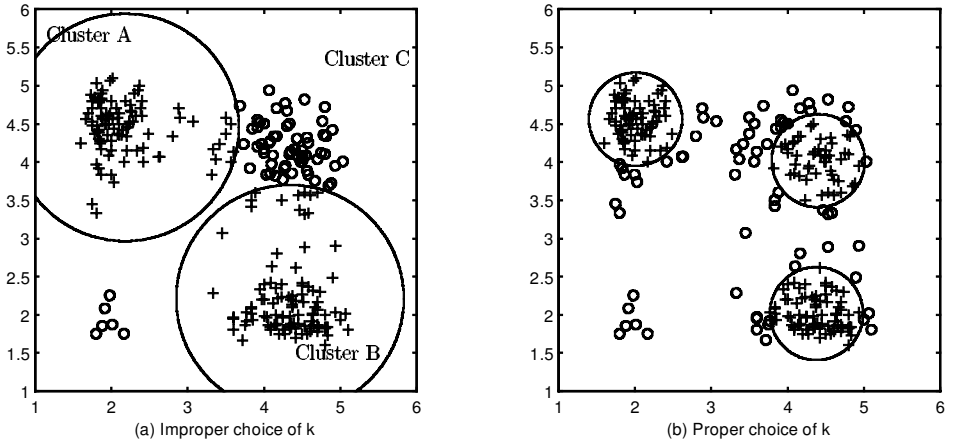


Figure 3. Bivariate "Old Faithful Geyser" dataset and optimal sets when $k = 2$ (a) and $k = 2$ (b) and $\alpha = 0.25$. Trimmed points: "$\circ$" symbols.

center of the distribution). Therefore, easy calculations yields the second derivative, W_1'', to be

$$W_1''(\alpha) = \frac{2}{(1-\alpha)^2} W_1(\alpha) + \frac{F^{-1}(1-\alpha/2) - F^{-1}(\alpha/2)}{2(1-\alpha)} \cdot \left[\frac{1}{f(F^{-1}(1-\alpha/2))} - \frac{F^{-1}(1-\alpha/2) - F^{-1}(\alpha/2)}{1-\alpha} \right].$$

As $W_1(\alpha)$ and $F^{-1}(1-\alpha/2) - F^{-1}(\alpha/2)$ are always positive, if the inequality

$$(F^{-1}(1-\alpha/2) - F^{-1}(\alpha/2)) f(F^{-1}(1-\alpha/2)) < 1-\alpha \tag{2.1}$$

holds, then W_1'' will be also positive. But, under our assumptions, inequality (2.1) is true (notice that the area of the rectangle in Figure 2 coincides with the left-hand side term of (2.1) and this area is clearly less than $1-\alpha$). So, sign changes for W_1'' are no longer possible. This fact also applies for general unimodal distributions with a slightly more complicated proof.

Given a random sample from distribution F, the $W_k(\cdot)$ functional can be studied through its plug-in estimator, the *empirical trimmed k-variance functional*, given by

$$\widehat{W_k} : \alpha \mapsto \widehat{W_k}(\alpha), \quad k = 1, 2, \ldots$$

where $\widehat{W_k}$ was defined in (1.3). For a continuous distribution function F, Theorem 3.6 in Cuesta-Albertos et al. (1997) establishes the almost sure convergence

$$\widehat{W_k}(\alpha) \to W_k(\alpha), \ F \quad \text{almost everywhere for} \quad \alpha > 0.$$

This convergence justifies the use of the empirical trimmed k-variance for estimating the analogous population statistic. In the remainder of this section, we will focus exclusively on this sample version.

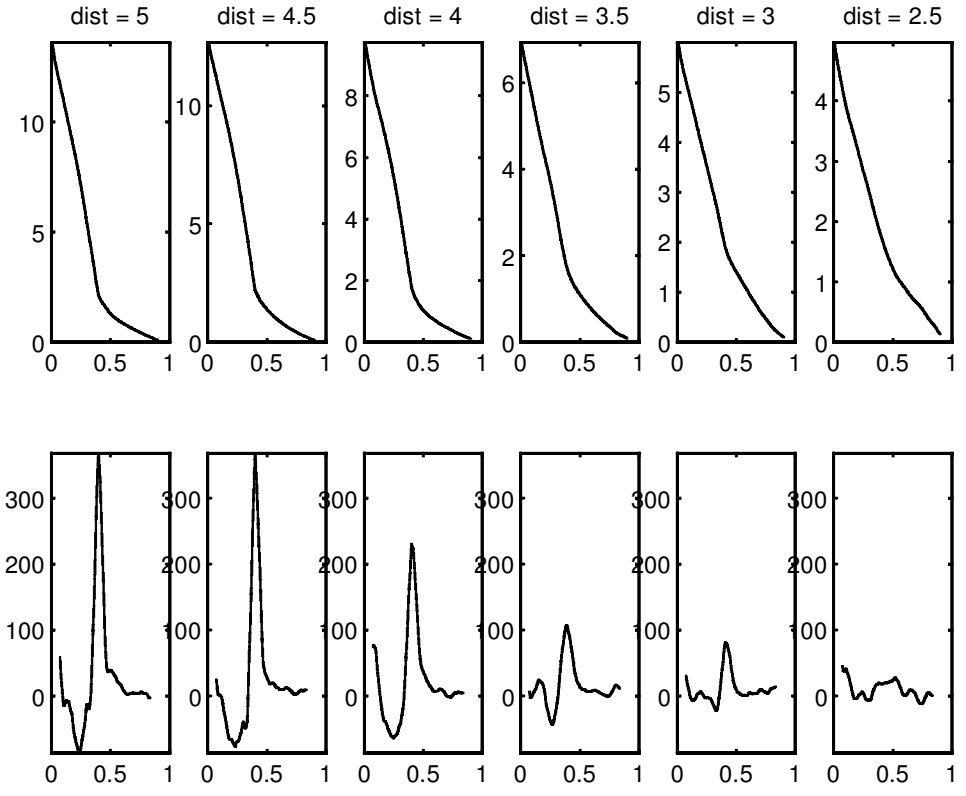


Figure 4. Curves $\widehat{W_1}$ *and numerical* $\widehat{W_1}''$ *for mixtures in Example 3. Parameter dist measures the distance of centers of clusters to* $(0,0)'$.

Example 2. The "Old Faithful" Geyser dataset contains 272 observations on the eruption lengths of that geyser (see, e.g., Azzalini and Bowman 1990). A bivariate dataset can be constructed considering the eruption lengths and the corresponding previous eruption lengths. Figure 3 shows this bivariate dataset, with three main clusters, and the trimmed k-means optimal zones for $k = 2$ and 3 for the same trimming size $\alpha = 0.25$. We can see how the "within groups" dispersion inside the optimal zone is greater when $k = 2$ (improper choice of k) than when $k = 3$ (proper choice). In Figure 3(a) we split unnaturally the main cluster C, into the other two clusters, A and B. The effect of trimming an additional data point in this figure (i.e., increasing slightly the trimming size) is to eliminate one of these artificially assigned points. This deleted observation had induced a large value in expression (1.3) and, therefore, the fact of trimming it reduces (1.3) notably.

Example 3. The mixture considered in Example 1 was rather simple because the two clusters there were well separated in order to exhibit a clear example. However, we could analyze more complicated cases if we bring the two clusters nearer. So, we sample now 90 observations from a $N(\mu_1, I)$ and 60 observations from a $N(\mu_2, I)$, where $\mu_1 = (0, \text{dist})'$, $\mu_2 = (\text{dist}, 0)'$ and dist ranging from 5 to 2.5. We can see in Figure 4 how the peak in the

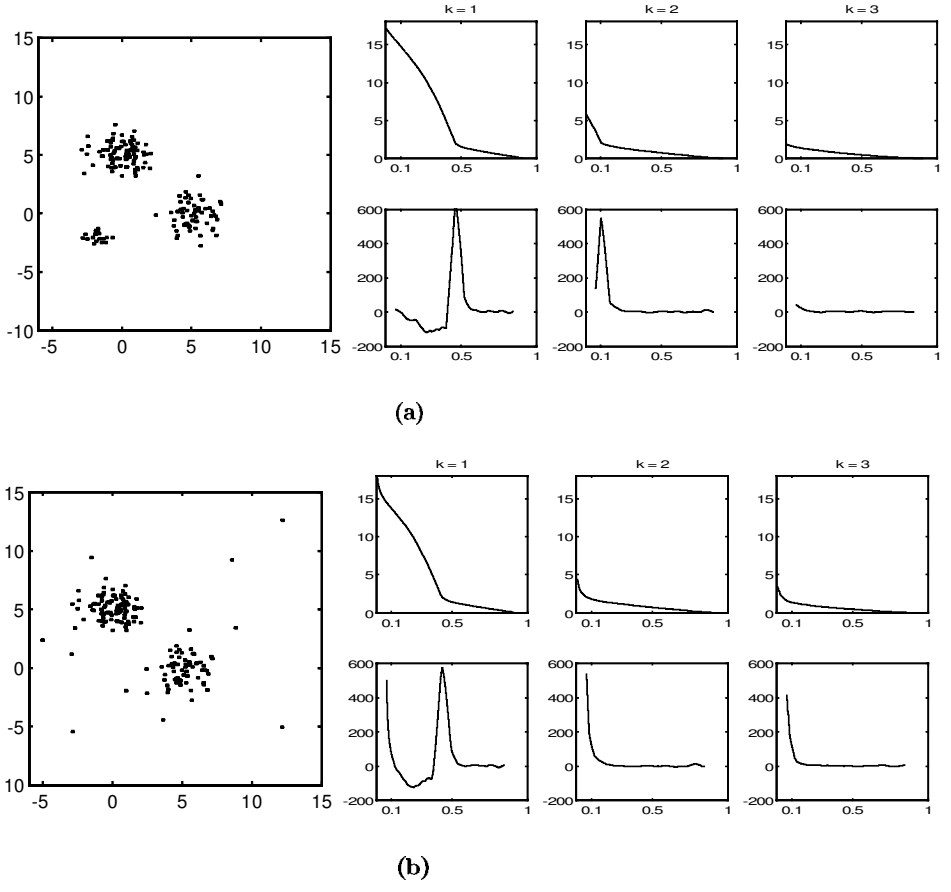


Figure 5. Datasets in Example 4 and their curves: Outlying group (a) and radial outliers (b).

second derivative at $\alpha = 0.4$ is made less marked when dist becomes smaller. However, we are able to detect this peak apart from case dist $= 2.5$. But, if we plot the mixture in that case we can see how these clusters are so close that it is practically impossible to detect them as a two-cluster structure.

Example 4. Often, the method is able to differentiate between two kind of outliers:

- *Outlying groups:* Figure 5(a) contains a dataset made up of 150 observations obtained in a similar fashion as in Example 1 and 17 data points (i.e., a group containing approximately 10% of the total mass) added with mean $\mu = (-2, -2)'$ and less scattered than the other two groups. Looking at curves in this figure, we observe a smooth decay when $k = 3$, suggesting the presence of three groups. However, the peak at $\alpha = 0.1$ in the $k = 2$ derivative curve tells us that the smaller group accounts only for 10% of the mass. The user should decide, with this information, whether this third group is a main cluster constituting part of the bulk of the data or,

on the contrary, it is merely an outlying group.

- *Radial outliers:* In the previous dataset, remove those 17 added observations and replace them by 17 radial outliers centered at $\mu = (3/2, 3/2)'$ and highly dispersed. The dataset and the result of applying the proposed method appears in Figure 5(b). The situation is different because a high k would be needed in order to obtain a smooth decay in all the range of α (recall that every radial outlier may be considered as a cluster itself). So, this kind of contamination can be detected when curves start by decreasing abruptly and this fact cannot be corrected unless we consider a notably high k.

2.1 TWO CASE STUDIES

2.1.1 Old Faithful Geyser

Let us consider again the "Old Faithful" Geyser data in Example 2. A bivariate dataset was defined there and Figure 6(a) shows its associated curves. These curves suggest $k = 3$ as a good choice for k. The presence of six "short followed by short" eruptions (data points in the left-down corner in Figure 3) produces an initial fast decrease in the trimmed k-variance functionals. These points do not induce a peak in the second derivatives because of the way that this numerical differentiation has been made (see comments about $\widehat{W_k}''$ in Section 3.2). But, from these curves, it is rather clear that these points do not constitute the bulk of data.

The dataset may be extended to a trivariate dataset considering the duration of eruptions and its first two lagged eruptions. If L denotes a long duration and S a short one, the situations LSL, SLS, LLL, SLL, LLS, LSS, SSL and SSS appear in the following percentages 31.5%, 18.9%, 16.3%, 14.4%, 14.4%, 2.2%, 2.2% and 0%, respectively. Now, in Figure 6(b), the smooth decrease in most of the range happens when we consider $k = 5$, coinciding with the bigger groups where sequences SS do not occur. Taking into account these curves, we can assert that the proportion of outliers is smaller than 10%, and therefore, applying the trimmed k-means clustering method for $k = 5$ and $\alpha = 0.1$ we could obtain the centers of the main groups without being affected by (rare) short followed by short eruptions.

Finally, notice that trimming levels that allow us to trim entire groups yield to peaks in the second derivative curve. So, the use of the trimming levels for which the last main peak is attained in each curve, together with the knowledge of the contamination level, could also lead to the obtention of an approximate determination of the sizes of the main groups.

2.1.2 Swiss Bank Notes

In order to show the performance of the procedure in a higher dimensional dataset, we analyze now the Swiss Bank Notes data (Flury and Riedwyl 1988). There, six variables are measured on 100 genuine and 100 forged bank notes, obtaining 200 data points in $\mathbb{R}^6$. Figure 7 shows the trimmed k-variance functionals and their numerical second derivatives for this dataset. We see in curves when $k = 1$ how the procedure clearly recognizes two

main clusters of similar sizes (genuine and forged bills).

However, we also observe that when $k = 2$ the second derivative curve does not reach stability until α is about 0.15 (even a small peak can be detected in that curve suggesting that a small group is being expelled). At this point, recall that the procedure is not only useful to compare how well represented the data may be through 2,3,. . . groups. It also tells us, as additional information, which elements are not "comfortable" in that representation. The two groups are clearly defined after trimming 28 bills ($\alpha = 0.14$) among which we can find precisely the set of 16 bills often detected as a third group in the literature. This third group is made up of 15 forged bills and one genuine, but misclassified, bill and it can

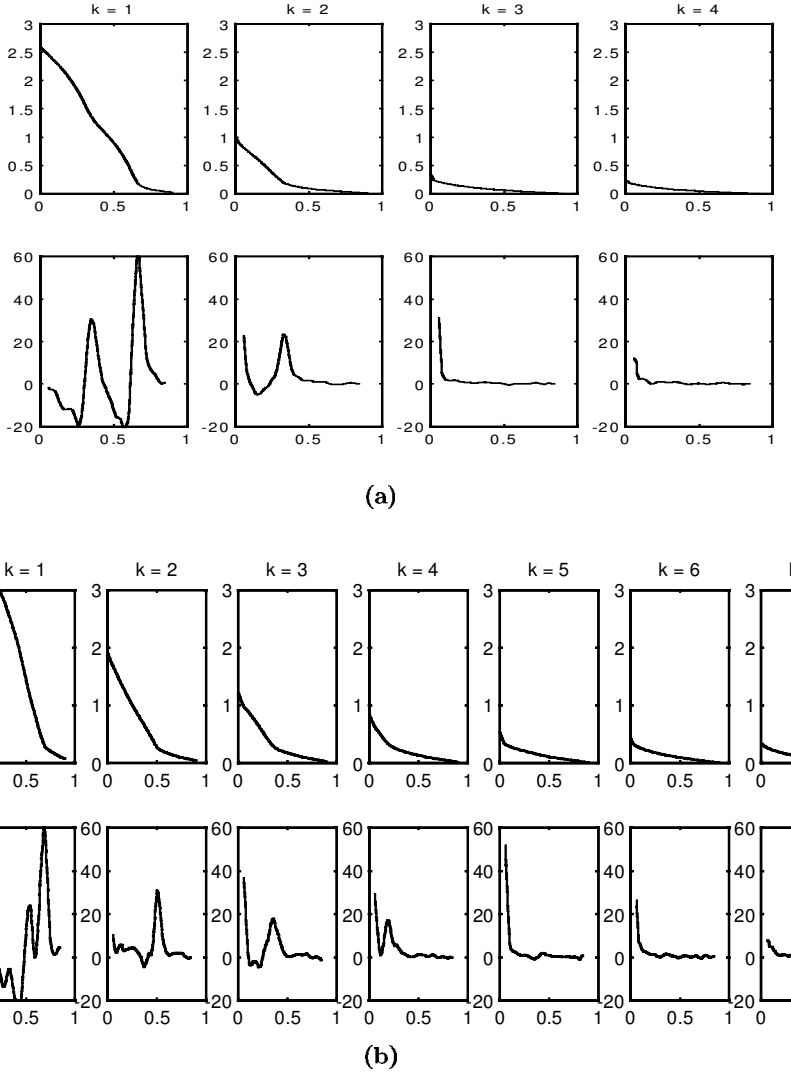


Figure 6. Curves for the bivariate (a) and the trivariate (b) "Old Faithful Geyser."

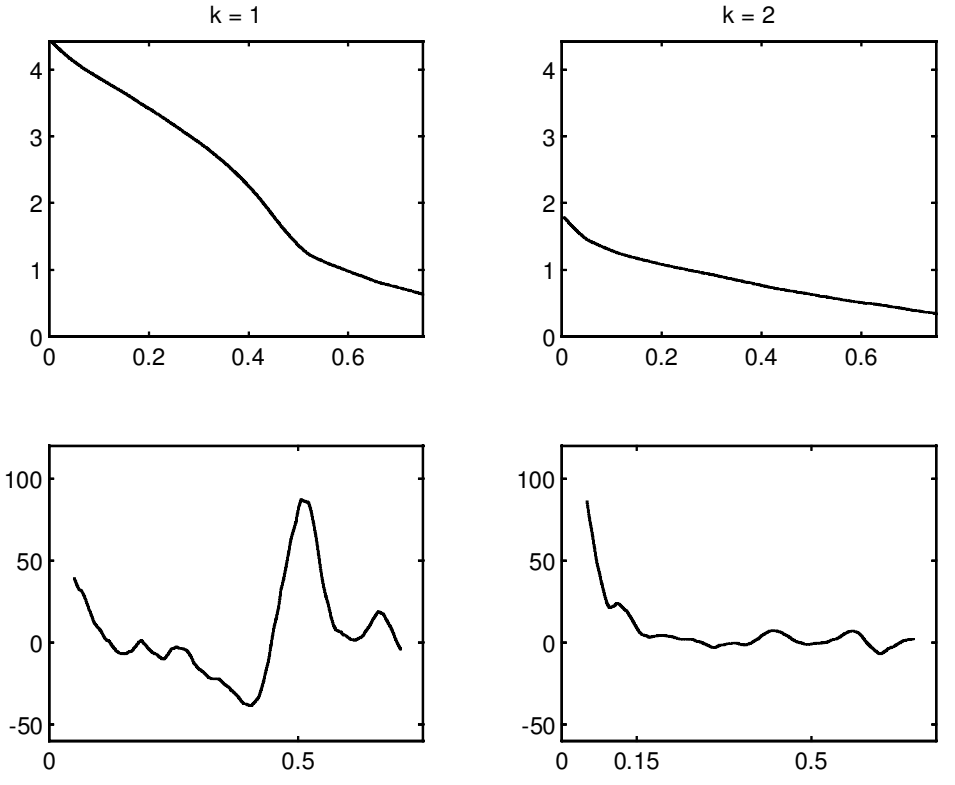


Figure 7. Curves for the "Swiss Bank Notes" data.

be obtained using standard multivariate techniques (see, e.g., Flury and Riedwyl 1988) or more sophisticated ones (e.g., Cook 1999).

That set of 28 trimmed bills contains the previously commented third group plus other remote observations with respect to the centers of the main groups. A careful analysis of that "residual" data using the same trimming techniques would show how those remote observations are sequentially trimmed off and a group essentially equal to that third group remains.

We would like to remark that this "third group" does not have the same importance as the two (main) groups and that it does not justify, for instance, the necessity of using k-means with $k = 3$. The 3-means clustering method does not create this "third group" as a group by itself and it prefers to break the forged bills into two other different groups (the same happens with Ward's clustering method). With trimmed 3-means, even trimming outlying observations, a partition of forged bills similar to the untrimmed one is obtained. However, in both cases ($k = 2$ and 3) our method detects that the "third group" is not "comfortable" within the main group of forged bills and it must be trimmed off in order to stabilize the second derivative curves.

3. COMPUTATIONAL ASPECTS

Trimmed k-means and the graphical methods proposed have obviously a high computational complexity, because dealing within the combinatorial space of subsets of a given dataset is needed. Exact algorithms are, in general, no longer possible and algorithm will be as important as the procedure itself.

The algorithm that will be proposed is a modification from the FAST-MCD algorithm in Rousseeuw and van Driessen (1999) for computing the minimum covariance determinant (MCD) estimator. The key feature will be the replacement of the, so-called, "C-step" by a "K-mean-step." In the C-step we keep the observations with lowest Mahalanobis distance from the last solution, and then a new solution based on those closest points is computed. Now, given k centers (last-solution), the new k centers are based only on the closest points by Euclidean distance to these centers.

3.1 ALGORITHM FOR COMPUTING TRIMMED k-MEANS

Given a dataset $\{x_1, \ldots, x_n\}$ and a trimming size α, the algorithm is as follows:

1. Select k starting points that will serve as seed centers (e.g., draw at random k observations from the whole dataset).
2. K-mean-step:
   Assume that $m_1, \ldots, m_k$ are the k centers obtained in the previous iteration:

   2.1. Compute distances of each observation to its nearest center

   $$d_i = \min_{j=1,\ldots,k} \|x_i - m_j\|, \quad i = 1, \ldots, n,$$

   and keep the set H having the $\lceil n(1-\alpha) \rceil$ observations with lowest d_i's.

   2.2. Split H into $H = \{H_1, \ldots, H_k\}$ where the points in H_j are those closer to m_j than to any of the other centers.

   2.3. The center m_j for the next iteration will be the mean of observations belonging to group H_j.
3. Repeat the K-mean-step a few times. After these iterations, compute the final evaluation function

$$\frac{1}{\lceil n(1-\alpha) \rceil} \sum_{j=1}^{k} \sum_{x_i \in H_k} \|x_i - m_j\|^2 . \tag{3.1}$$

4. Draw random starting centers (i.e., start from step 1) several times, keep the solutions leading to minimal values of the evaluation function (3.1) and fully iterate them to choose the best one.

Notice that the K-mean-step is similar to iterations of the classical algorithms for computing k-means (see, e.g., McQueen 1967), but we retain only the proportion $1 - \alpha$ of closer

observations instead of all the observations. It is not difficult to show that the solution obtained after a K-mean step is at least as good as the previous one. So, each sequence of iterations should converge to a local minima and several reinitializations are used trying to attain the global minima.

3.2 ALGORITHM FOR COMPUTING TRIMMED k-VARIANCE FUNCTIONALS

The previous trimmed k-means algorithm can be now used for obtaining the trimmed k-variance functionals. If K is the total number of centers to be examined, we construct for $k = 1$ to K a grid of increasing trimming sizes $\{\alpha_i\}_{i=1}^{I}$ partitioning the interval $[0, 1]$. Then, we carry out a trimmed k-mean algorithm to obtain $\widehat{W_k}(\alpha_i)$ for each α_i.

The total number of groups to be analyzed, K, is settled in an interactive way if no initial value is available. We stop at the first k such that the decay of $\widehat{W_k}$ becomes soft in most of its range. Also, the choice of an equally spaced grid, $\alpha_i = i/(I+1)$, is preferable to compute the numerical derivatives.

The most expensive part of the algorithm are the trimmed k-means computations, and some ideas for speeding up this step are:

- If we begin with good starting seeds, few K-mean-steps are needed. Although improvements in position of centers can be obtained, we have observed that the magnitude of $\widehat{W_k}$ is soon close to the optimal one.
- The choice of the number of random initializations is surely the keystone of the performance of the algorithm. Assuming that there are exactly k clusters with similar weights and a contamination level α, it is easy to see that the probability of having one center seed in each group among m initializations is $1 - \left(1 - k!\left(\frac{1-\alpha}{k}\right)^k\right)^m$. So, if we desire that this probability exceed p we need at least

$$m \geq \frac{\log(1-p)}{\log\left(1 - k!\left(\frac{1-\alpha}{k}\right)^k\right)} \tag{3.2}$$

  initializations.
- To obtain $\widehat{W_k}(\alpha_i)$ it is convenient to impose one nonrandom initialization with seeds corresponding to the empirical trimmed k-means obtained in the previous iteration for trimming size α_{i-1}. This initialization leads at least to the reduction of $\widehat{W_k}$ that the simple increase of the trimming size will produce if no great changes in the location of centers happen at this trimming size.
- Formula (3.2) could lead to a considerable number of random initializations if k and α are great. However, notice that in this case it is easy to have already surpassed the trimming level where the fast decay stops. In other words, k could be greater than the number of groups remaining after the trimming process. This fact produces high instability due to very similar (and not very interesting) solutions. Therefore, this case requires less precision and lesser number of initializations. To make precise

this claim, suppose we make use of $\widehat{W_k}$ to detect sequentially the presence of at least $k+1$ groups, $k = 1, 2, \ldots$ (with $\widehat{W_1}$ we detect the presence of at least 2 groups, and so on). If we have $k+1$ or more groups in our data, then the smallest group should have less than $1/(k+1)$ of mass. Thus, we must notice on $\widehat{W_k}$ the effect of eliminating this smallest group before reaching this trimming size. So, we only need a precise resolution in $\widehat{W_k}$ when α ranges in the interval $(0, 1/(k+1)]$.

- If the equispaced grid was taken, a second derivative approximation, $\widehat{W_k}''(\alpha_i)$ may be obtained as $((I+1)/h)^2(\widehat{W_k}(\alpha_{i-h}) - 2\widehat{W_k}(\alpha_i) + \widehat{W_k}(\alpha_{i+h}))$ where $h \in \mathbb{N}$ is a parameter that controls the roughness of the numerical second derivative.
  In some sense, parameter h plays an analogous role to that of the bandwidth parameter in density estimation. If h is small, functionals are rough and more data dependent. However, if h is too high the resolution is diminished and we cannot detect features of size smaller than h. In the bivariate "Old Faithful Geyser" data, h had been chosen greater than the size of the smallest group. So, a peak in the numerical derivative due to this small group does not appear. However, we would like to point out that the detection of bigger groups constituting the bulk of data is not very dependent on the particular choice of h.

The implementation of algorithms leading to figures in this paper takes some minutes in most of the cases. A MATLAB code for doing these procedures is available at http:/www.est.cie.uva.es/~langel/software. This code is not fully developed and many improvement (taking into account the above suggestions) are still in progress to obtain better computing times and efficiency.

4. MIXTURES OF ELLIPTICAL DISTRIBUTIONS

A clear weakness of the proposed approach is that it is specially aimed to mixtures of spherical distributions. This is a shared problem with k-means and other techniques based on this clustering method (notice that (1.1) does not take into account the particular shape of each cluster in the mixture). Although in many cases the proposed simple approach serves perfectly to detect clusters and outliers, it is obvious the improvement that allowing for different covariance matrices in each group could have.

A possible solution is based on replacing the curves based on $\widehat{W_k}$ by curves obtained from the problem

$$\min \sum_{i=1}^{k} n_i \log(\det(W_i/n_i)), \tag{4.1}$$

where n_i and W_i denote, the size and a scatter estimation matrix of the non-trimmed observations within group i, and where a proportion α of trimmed observations is allowed [details concerning this robust classification method can be found in Rocke and Woodruff (1999)].

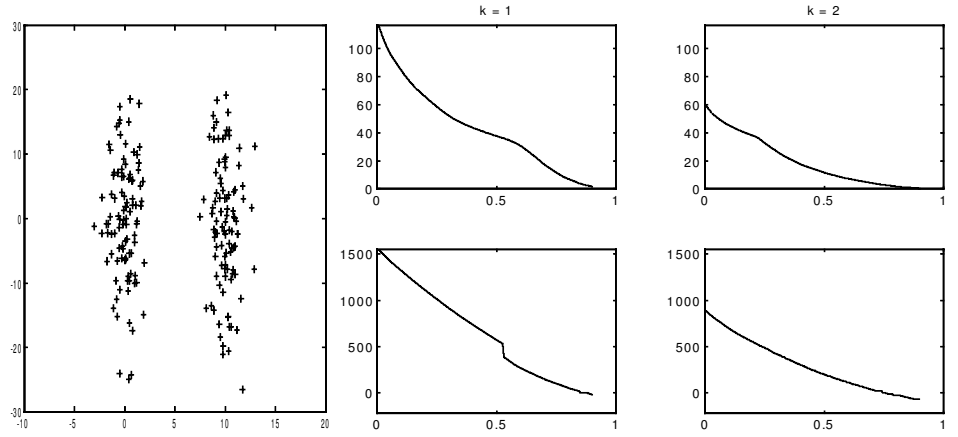


Figure 8. Trimmed k-variances ($k = 1$ and 2) curves (top) and the same curves (bottom) based on (4.1) for clusters in Section 4.

Figure 8 (top) exhibits the trimmed k-variance functionals when $k = 1$ and 2 for a sample composed of 100 observations from a bivariate normal with mean vector $(0, 0)'$ and 100 observations from another bivariate normal with mean vector $(10, 0)'$, and both having covariance matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 100 \end{pmatrix}.$$

Curves $\widehat{W_1}$ and $\widehat{W_2}$ do not allow us to detect properly the two-clusters structure (notice where changes in the rates of decrease happen) for these highly linear clusters (far away from being spherical).

However, in Figure 8 (bottom) we see a clear "jump" at $\alpha = 0.5$ when $k = 1$ in the curve obtained plotting the values obtained from (4.1) against α (discovering that each group accounts for 50% of the data points).

Notice that we could replace (4.1) by the result of other ways of measuring the trimmed discrepancy between a set of k specially chosen points in the sample space and the whole dataset. We believe that this methodology could be interesting, because it allows the consideration of "metrics" specially designed for each problem. We are currently working in this direction.

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